FAIZ SIDDIQ

Solution Architect | GenAI and Agentic AI Engineer | Backend and Cloud (AWS) | Technical Pre-Sales

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SUMMARY

Solution architect and GenAI engineer with 6+ years of experience, including 5 years at DvSum building agentic AI, data platform, and hybrid cloud systems for telecom operators such as Liberty Latin America (LLA), Rogers, and izzi. Work across the full DvSum stack: a serverless AWS backend of about 70 CDK stacks and 130+ Lambda functions, the Java 21 on-premises gateway that connects to 20+ enterprise data sources, and the LangChain and Bedrock chat agent. Designed, built, and later refactored DvSum's AI Agents end to end. Own LLA delivery from requirements and architecture through build, QA, production release, and customer success, and run technical demos and POCs that turn into approved projects.

SKILLS

Languages: Python, Java 21, Go, SQL (PostgreSQL, PL/pgSQL), JavaScript (Node.js), Shell

GenAI and Agents: Agentic AI, ReAct agents, multi-agent routing (intent, chain, and tool selection), tool and function calling, Model Context Protocol (MCP), RAG with pgvector, human-in-the-loop workflows, prompt engineering, token and latency optimization, LLM-based PII classification and masking, multilingual (English and Spanish) responses

LLM Platforms: AWS Bedrock (Claude Sonnet, Claude Haiku), Anthropic API, OpenAI, Azure OpenAI, LangChain, tiktoken

Backend: Spring Boot 3, Flask, FastAPI, Django, REST APIs, WebSockets, JSON and YAML workflow engines, Jinja2 templating, multithreading and concurrency control

AWS: Lambda, Step Functions, API Gateway (REST and WebSocket), EventBridge, ECS Fargate, ECR, EC2, S3, Aurora PostgreSQL, DynamoDB and Streams, ElastiCache Redis, SQS, SNS, Kinesis, DMS, Cognito, IAM, KMS, Secrets Manager, SSM Parameter Store, CloudWatch, PrivateLink, Site-to-Site VPN, Route 53, ALB

Azure: App Services, Functions, API Management, Blob Storage, ADLS, Azure SQL, Synapse Analytics, Microsoft Fabric and OneLake, Event Grid, Key Vault, Cosmos DB, Azure DevOps

Data Sources: Oracle, SQL Server, PostgreSQL, MySQL, Snowflake, Databricks, DB2, Netezza, Azure Synapse, Amazon S3, REST and SOAP APIs, DuckDB, Redis Streams

Integrations: Jira, ServiceNow, Microsoft Teams, Twilio, WhatsApp, SFTP, Airtable, SharePoint, Power BI

DevOps and Testing: AWS CDK, Terraform, Docker, Kubernetes, Helm, Gradle, Git, Jenkins, CI/CD, Linux, systemd, pytest, moto, JUnit, load testing

Delivery: Requirements gathering, solution architecture, technical pre-sales, POCs, stakeholder workshops, QA and UAT, production support, code review, Agile and Scrum, technical documentation

EXPERIENCE

DvSum (Data Intelligence Platform), Remote (California, USA)

Senior Backend Engineer and Solution Architect | Nov 2021 - Present

Promoted from Backend Engineer to Senior Software Engineer, then Associate Cloud Architect, then current role.

- Work across DvSum's three core codebases: the serverless backend (about 70 CloudFormation stacks, 130+ Lambdas, and 40 product domains on one shared layer), the DvSum Gateway (Java 21, Spring Boot 3.5, Python sidecar), and the chat agent (LangChain, Bedrock, pgvector).
- Designed and built DvSum's AI Agent platform from scratch, then refactored it into a config-driven system that runs in new markets without code changes. The one engineer with end-to-end knowledge of how the agents work.
- Own all Liberty Latin America work: requirements, design, build, pre- and post-release QA, production releases, and customer success calls with LLA support teams and end users. Lead the engineering team on LLA deliveries.
- Delivered for other telecom customers too: fixed production gateway issues for Rogers (log flooding, disk growth from scan files, log rotation) and cut LLM response time for izzi's chat agent by more than half.
- Enforce platform standards in design and code review: tenant identity only from the authorizer, cross-tenant requests returning 404, whitelist-built responses, additive-only changes to shared stacks, and new features shipped behind flags that default to off.
- Set up private connectivity for customer networks with AWS Site-to-Site VPN and PrivateLink, and contributed to data catalog features, including Microsoft Fabric and OneLake scanning and LLM-based Power BI lineage extraction.
- Run technical demos and POCs for new projects. The Bahamas auto ticketing POC was approved for production after phase one.

Key projects at DvSum

AURA Agentic AI Orchestrator | Architect and Backend Engineer

- Runtime where every chat session gets its own short-lived ECS Fargate task, running a ReAct loop with Claude on Bedrock and in-process tools for loading S3 data, DuckDB queries, Python, and shell.
- A spawner Lambda, a DynamoDB session registry, and API Gateway WebSockets route follow-up messages to the live session, with each session's workspace stored on S3.
- Moved the four agentic stacks into the main CDK app as in-place CloudFormation updates with no resource recreation, and wired in PrivateLink and per-account private routing using the existing platform pattern.
- Porting the runtime from Python to Go, checked against golden fixtures captured from real sessions. Wrote telecom domain skills for FTTH and OLT, RAN mobile, and tech support troubleshooting.

  Tech: Python, Go, AWS Bedrock (Claude), ECS Fargate, Lambda, DynamoDB, API Gateway WebSocket, DuckDB, S3, AWS CDK

AURA Workflow Orchestration Engine | Backend Engineer

- Step-based workflow engine with 20 step types, including AI agent, LLM call, API, database, code, decision, fan-out, for-each, child workflow, human approval, Jira, email, SFTP, and monitor steps.
- Decision steps evaluate Jinja2 switch and expression rules, human-in-the-loop steps collect approvals over email and Microsoft Teams, and AI steps can run on Bedrock, OpenAI, or Azure OpenAI.
- Runs on SQS, DynamoDB Streams, and Lambda, with execution locks, cancel and stop guards, and failure continuation so long workflows recover cleanly.

  Tech: Python, AWS Lambda, SQS, DynamoDB Streams, Aurora PostgreSQL, Jinja2, Twilio, Microsoft Teams

DvSum Chat Agent and LLM Optimization | Lead Developer

- LangChain multi-agent chat service with chain selection, tool selection, and action agents, an MCP tool layer for API, JDBC, and custom Python function tools, and RAG retrieval with pgvector.
- Built LLM-based sensitive-data masking that sends only field names and value-free hints to the model, never the values. Added English and Spanish translation and a custom spaCy NER model for parsing commands.
- Cut chat agent latency for izzi by having the LLM write only the narrative while code injects the structured data. Output tokens dropped from 2,658 to about 900 (66% fewer) and LLM time from 41 to 16 seconds (62% faster), with end-to-end response time estimated to drop from 91 to 66 seconds.
- End-to-end tests showed 54% less chain time in English and 46% in Spanish. Added custom Python function execution to action chains across all three services without changing existing flows.

  Tech: Python, LangChain, AWS Bedrock (Claude Haiku and Sonnet), Anthropic API, OpenAI, pgvector, Redis, FastAPI, Flask, spaCy

Proactive Network Management (Proactive Workflows) with Smart Scan | Architect and Backend Lead

- Backend for DvSum's proactive workflows, an AWS Step Functions pipeline with four Lambda stages that pulls KPI data for fiber nodes through the on-prem gateway, analyzes each node, and raises incidents with a root cause and next step. Designed to flag degradation hours before customers call, with a P99 target of under 90 seconds per incident.
- Designed Smart Scan so the pipeline only works where something changed. Nodes are grouped into cadence buckets by priority and classified each run as FULL, RESCOPE, DELTA, SKIP, or RESOLVED. Critical nodes are still checked every 15 minutes, but unchanged nodes run 0 queries. In design simulations, about 40% of nodes skipped querying on a typical run.
- Moving lower-priority nodes from a 15-minute to an hourly cadence cuts their scans by 75%. On the gateway side, node queries run through a tiered priority queue with a single database concurrency gate, so the customer's source is never overloaded.
- Capped LLM use at 2 calls per incident against 3 to 15+ for an open-ended agent, with prompts of about 1,350 tokens built from SQL summaries. This puts the target cost at $0.02 to $0.05 per incident, 70 to 95% lower than the agent approach.
- Made runs finish every time with a watchdog, exactly-once completion through Redis, and per-node guards in DynamoDB. Found and fixed a silent defect that would have forced a full rescan of every node on every run.
- Shipped with 1,200+ automated tests, including a 261-test legacy regression suite. Wrote the API contract (32 verified cases) for the frontend team, worked with the DB team on 7 stored procedures, and wrote the QA guides used to onboard the automation team.

  Tech: Python, Java, AWS Step Functions, Lambda, EventBridge, DynamoDB, Redis, Aurora PostgreSQL, SSM, AWS CDK

DvSum Gateway: High Availability and Production Reliability | Architect and Developer

- Designed high availability for the on-prem gateway on Kubernetes with Helm: a persistent gateway tier plus short-lived job pods, Kubernetes Lease leader election, and a Redis Streams and Sentinel job queue that moves poison jobs to a dead-letter queue after 3 attempts.
- Fixed a production problem where workflow actions timed out in bulk while CPU and memory looked healthy. The cause was thread-pool self-exhaustion plus missing timeouts. Split the pools and added timeouts, with 1,165 of 1,165 tests passing, load tests included.
- Maintain connectors for 20+ data sources, including Oracle, SQL Server, Snowflake, Databricks, DB2, Netezza, Azure Synapse, Microsoft Fabric, Amazon S3, REST, and SOAP.

  Tech: Java 21, Spring Boot 3.5, Python, Kubernetes, Helm, Redis Streams, Redis Sentinel, Docker, Gradle, JUnit

AI Troubleshooting Agent for HFC and FTTH, Liberty Latin America | Lead Architect

- Built a JSON-driven workflow engine that fetches, filters, and processes data inside LLA's network, so diagnostics run without moving raw data out.
- The agent works through signal issues, billing problems, provisioning gaps, and outages step by step, and gives Tier-2 support a root cause instead of another ticket.
- Ran 60+ logic design and review sessions with Tier-2 leads for FTTH and HFC. Live in 5+ markets, including Jamaica, Saint Lucia, Trinidad, and Grenada.
- Added audience-specific LLM prompt flows with output as plain text, Markdown, or JSON.

  Tech: Python, Java (Spring Boot), OpenAI API, AWS Lambda, API Gateway, REST APIs

WhatsApp AI Agent Integration, Liberty Latin America | Project Lead and Backend Developer

- Connected the DvSum AI Agent to LLA's customer-facing WhatsApp bot so customers can troubleshoot their own service.
- Replaced market-specific code with payload-based controls, so one deployment serves 8+ markets.
- Designed three response levels from the same diagnosis: technical for support staff, simplified for frontline agents, and plain language for customers.
- Coordinated 3 external teams through requirements, API documentation, and workshops. Built the backend myself and support it in production.

  Tech: Python, Java (Spring Boot), OpenAI API, WebSockets, AWS Lambda

Bahamas Proactive Auto Ticketing, Liberty Latin America | Cloud Architect and Project Manager

- Led the POC, delivered it on time, and got the project approved for production.
- Built a daily pipeline on ECS Fargate, EventBridge, and S3, set up with AWS CDK and Terraform, that finds FTTH faults and books technician visits before customers call. Reports go to SharePoint to refresh Power BI dashboards.
- Ran daily acceptance testing with tech support and tuned the logic until false positives reached zero, which stopped unnecessary technician visits.

  Tech: Python, AWS CDK, Terraform, ECS Fargate, EventBridge, S3, SharePoint, Power BI

Azie Technologies, Remote

Freelance Mobile Developer (Flutter, Android) | Mar 2020 - Nov 2021

- Built and shipped Flutter and Firebase apps for clients, including DevPay Analytics, a Stripe-based dashboard for merchants.

EDUCATION

BS Software Engineering, COMSATS University Islamabad | Sep 2017 - Aug 2021

CERTIFICATIONS

Certified Scrum Agile Master (CASM), HRSD | Verify credential